Experiment No.2

Title: To refactor the Python script to use CSS selectors and XPath for data extraction.

Batch: B3 **Roll No.: 16014225801** **Experiment No.:2**

Aim: To perform web scraping on a dynamic website using Python libraries such as Selenium and BeautifulSoup, and to extract data that loads asynchronously through JavaScript.

Resources needed: Internet

Theory:

Dynamic websites generate or update content on the client side through JavaScript , AJAX , or API calls after the initial HTML page is loaded . Unlike static pages (where the required data is available in the HTML source), dynamic pages load content asynchronously, meaning that simply requesting the page's HTML won't contain the desired data.

To scrape such websites, we use browser automation tools like Selenium, which can execute JavaScript, render the full page, and expose the final HTML to be parsed. Selenium simulates user interactions such as scrolling, clicking, or waiting for specific elements to load, enabling the extraction of dynamically generated data.

Once the rendered page source is obtained, libraries like BeautifulSoup are used to parse the HTML and extract relevant information.

Procedure:

Step 1: Install the lxml Library

Step 2: Refactor with CSS Selectors

Step 3: Refactor with XPath

Results:

```
# Dynamic Web Scraping using Selenium and BeautifulSoup

from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.chrome.options import Options
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from bs4 import BeautifulSoup
import time

# 1. Configure Selenium
options = Options()
options.add_argument("--headless")
options.add_argument("--disable-gpu")
options.add_argument("--window-size=1920,1080")
driver = webdriver.Chrome(options=options)

# 2. Open Dynamic Website
url = "https://quotes.toscrape.com/js/"
driver.get(url)

WebDriverWait(driver, 10).until(EC.presence_of_element_located((By.CLASS_NAME,
"quote")))
print("Page loaded successfully")

# 3. Extract data using Selenium + CSS Selectors
```

```
quotes = driver.find_elements(By.CSS_SELECTOR, ".quote")
print("\n--- Using CSS Selectors ---")

for quote in quotes:
    text = quote.find_element(By.CSS_SELECTOR, ".text").text
    author = quote.find_element(By.CSS_SELECTOR, ".author").text
    print("Quote :", text)
    print("Author:", author)
    print()

# 4. Extract data using XPath
print("\n--- Using XPath ---")
quotes_xpath = driver.find_elements(
    By.XPATH, "//div[@class='quote']"
)

for quote in quotes_xpath:
    text = quote.find_element(
        By.XPATH, ".//span[@class='text']"
    ).text

    author = quote.find_element(
        By.XPATH, ".//small[@class='author']"
    ).text

    print("Quote :", text)
    print("Author:", author)
    print()

# 5. Get rendered HTML
html = driver.page_source

# 6. Parse rendered HTML using BeautifulSoup
soup = BeautifulSoup(html, "lxml")
print("\n--- Using BeautifulSoup ---")
quote_blocks = soup.select(".quote")
for block in quote_blocks:
    quote_text = block.select_one(".text").get_text(strip=True)
    author_name = block.select_one(".author").get_text(strip=True)
    print("Quote :", quote_text)
    print("Author:", author_name)
    print()

# 7. Close browser
driver.quit()
```

Outputs:

```
Page loaded successfully

--- Using CSS Selectors ---
Quote : “The world as we have created it is a process of our thinking. It cannot be
changed without changing our thinking.”
Author: Albert Einstein
```

```
Quote : “It is our choices, Harry, that show what we truly are, far more than our
abilities.”
Author: J.K. Rowling

Quote : “There are only two ways to live your life. One is as though nothing is a
miracle. The other is as though everything is a miracle.”
Author: Albert Einstein

Quote : “The person, be it gentleman or lady, who has not pleasure in a good novel,
must be intolerably stupid.”
Author: Jane Austen

Quote : “Imperfection is beauty, madness is genius and it's better to be absolutely
ridiculous than absolutely boring.”
Author: Marilyn Monroe

Quote : “Try not to become a man of success. Rather become a man of value.”
Author: Albert Einstein

Quote : “It is better to be hated for what you are than to be loved for what you are
not.”
Author: André Gide

Quote : “I have not failed. I've just found 10,000 ways that won't work.”
Author: Thomas A. Edison

Quote : “A woman is like a tea bag; you never know how strong it is until it's in hot
water.”
Author: Eleanor Roosevelt

Quote : “A day without sunshine is like, you know, night.”
Author: Steve Martin


--- Using XPath ---
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--- Using BeautifulSoup ---
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Author: Steve Martin
```

Conclusion:

In this experiment we successfully performed web scraping on a dynamic website using Python libraries such as Selenium and BeautifulSoup, and to extract data that loads asynchronously through JavaScript.

Grade: AA / AB / BB / BC / CC / CD /DD

Signature of faculty in-charge with date

References:

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2. Anish Chapagain, “Hands-On Web Scraping with Python Extract Quality Data from the Web Using Effective Python Techniques”, Packt Publishing, 2nd Edition 2023.
3. Pradumna Milind Panditrao, “A Python Guide for Web Scraping: Explore Python Tools, Web Scraping Technique”, BPB, 2nd Edition 2019.